

Lab One

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1 CRAFTING A COMPILERS 1.11

Investigate the techniques MOSS uses to find similarity. How does MOSS differ from other approaches for detecting possible plagiarism?

Other approaches tend to use a method where the document is divided into x-grams (user decides what x is). After each gram is hashed, a subset of the hashes becomes the fingerprints of the document. Then choose all hashes that are $0 \bmod p$ for some fixed p . The downside to this method is that it gives no guarantee that matches between documents are detected. The maximum gap between consecutive fingerprints is unbounded and any matches in these gaps are not detected. MOSS introduces a more efficient algorithm that guarantees at least any part of a long enough match is detected. They do this through windows, consecutive hashes of x-grams in a doc, The algorithm selects one fingerprint from every window and limits the maximum gap between fingerprints.

2 CRAFTING COMPILERS 3.1

What token sequence is produced? For which tokens must extra information be returned in addition to the token code?

```
T_qualifier_const > T_type_float > T_ID > T_assign > T_decimal > T_EOF
T_type_float > T_ID > T_EOF
T_type_int > T_ID > T_assign > T_digit > T_EOF
T_ID > T_assign > T_digit > T_EOF
T_while > T_left_paren > T_ID > T_greater_than > T_right_paren > T_left_bracket > T_EOF
T_print > T_left_paren > T_string_literal > T_args_seperator > T_ID > T_args_seperator T_ID >
T_right_paren > T_EOF
T_ID > T_assign > T_ID > T_minus_op > T_ID > T_plus_op > T_decimal > T_multiplication_op
> T_ID > T_EOF
T_ID > T_assign > T_plus_op > T_digit > T_EOF
```

Extra information needs to be returned for identifiers. Their values are important.

3 DRAGON BOOK 1.1.4

A compiler that translates a high level language into another high-level language is called a source to source translator. What advantages are there to using C as a target language for a compiler?

At this point, C is a universal language. Any device capable of running C can run the code loaded into the compiler. You can easily translate code from one language into C.

4 DRAGON BOOK 1.6.1

For the block-structured C code of Fig. 1.13(a), indicate the values assigned to w, x, y, and z.

```
w = 13,  
x = 11  
y = 13  
z = 11
```